Stage 1 Report — Macroeconomic Forecasting with Large Language Models

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Paper reproduced: Carriero, Pettenuzzo & Shekhar (2026), *"Macroeconomic Forecasting with Large Language Models,"* arXiv:2407.00890v5.

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1. The paper, in brief

Problem. Can Time Series Foundational Models (TSFMs) — pretrained transformer models for time series, the "LLMs of forecasting" — beat traditional econometric methods at forecasting macroeconomic variables?

Data. FRED-MD: 120 monthly US macro series (output, prices, labor, housing, money/credit, rates/FX, stock market), Jan 1959–Dec 2022, each series transformed to near-stationarity via McCracken & Ng (2016)'s standard codes (log-differences, first differences, etc.).

Models compared. - *Econometric:* AR(1) (mandatory benchmark), BVAR with a Minnesota-tradition Normal-Inverse-Wishart prior (p=6 lags, hyperparameters tuned by maximizing the marginal likelihood), a PCA-factor-augmented autoregression, NNAR, and BART. - *TSFMs (all zero-shot):* Chronos (Amazon), Moirai (Salesforce), TimesFM (Google), LagLlama, TTM (IBM), Time-GPT (Nixtla) — ranging 1M–311M parameters. - *Hybrid:* fit BVAR -> take its in-sample residuals -> forecast those residuals zero-shot with a TSFM -> add the two forecasts together (their "TSFM-BVAR").

Method. Recursive expanding-window forecasting: train on Jan 1960–Dec 1984, forecast Jan 1985 (h=1..12), roll forward one month, repeat through Dec 2022. Accuracy is measured as RMSFE relative to the AR(1) benchmark (their Eq. 11):

```
RMSFE_ijh = sqrt( Sum e^2_{i,j,tau+h} / Sum e^2_{AR1,j,tau+h} )
```

Values below 1 mean model *i* beats AR(1) for series *j* at horizon *h*.

Headline findings. 1. Only 3 of 6 TSFMs (Chronos, TimesFM, Moirai-large) are consistently competitive with AR(1); TTM, LagLlama, Time-GPT lag behind. 2. Even the good TSFMs are only roughly on par with BVAR/Factor models, not clearly better. 3. BVAR and TSFM forecast errors are nearly *orthogonal* at h=1 (correlation ~0.07–0.5) — they're wrong about different series — which is exactly why combining them (the hybrid) helps: it substantially trims the worst-case (max) errors of the TSFM. 4. TSFMs handle the COVID-19 shock better than econometric models (with the caveat that their training data may include the pandemic period).

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2. What we actually built (Stage 1 requires real execution, not description)

Full pipeline, all runnable from `/Users/nilesh/LLM`:

Script	Does
`scripts/prep_data.py`	Loads FRED-MD CSV, selects 60-series subset, applies the 7 McCracken-Ng transform codes, saves `data/fredmd_clean.parquet`
`scripts/econ_models.py`	AR(1), Minnesota-prior BVAR, PCA-factor-augmented AR — shared `_bvar_fit()` used by both the forecast function and the hybrid's residual extraction

Script	Does
`scripts/run_forecast_loop.py`	Recursive expanding-window loop + `rmsfe_table()` (paper's Eq. 11)
`scripts/run_tsfm_chronos.py`	Amazon Chronos (bolt-small, zero-shot)
`scripts/run_tsfm_timesfm.py`	Google TimesFM 2.5 (200M, zero-shot)
`scripts/run_tsfm_moirai.py`	Salesforce Moirai (base, 91M, zero-shot)
`scripts/run_tsfm_ttm.py`	IBM Tiny Time Mixer (5M, zero-shot)
`scripts/run_tsfm_lagllama.py`	LagLlama (25M, zero-shot)
`scripts/run_tsfm_timegpt.py`	Nixtla Time-GPT via API (batched: 1 call/origin covering all 60 series, not 1 call/series)
`scripts/run_hybrid.py`	TSFM-BVAR hybrid: BVAR forecast + TimesFM-modeled BVAR residuals
`scripts/merge_tsfm_errors.py`	Folds any model's raw forecasts into the shared error table
`scripts/make_plots.py`	Boxplots mirroring the paper's Figure 2/3

Environments: `venv_core` (Python 3.14 — torch, chronos-forecasting, timesfm, sklearn, statsmodels), `venv_moirai` and `venv_lagllama` (Python 3.11, forced by `uni2ts` / `granite-tsfm` pinning scipy/torch versions with no 3.14 wheel — the exact kind of dependency conflict the course guide warned about). All six TSFMs and both econometric benchmarks actually ran to completion on real data; nothing here is simulated or hand-waved.

Two real implementation bugs were found and fixed during development by comparing early results against the paper's expected direction (median RMSFE should hover near 1.0, not be systematically 1.3+): 1. **BVAR dummy-prior bug**: a lag-decay multiplier was accidentally a no-op (`lag if lag==1 else 1` always evaluates to 1), so the Minnesota prior wasn't actually shrinking higher lags more than lag 1. Fixed to scale properly as `lag * sigma_i / lambda` (paper's Eq. B.1, Var ~ lambda^2/lag^2). Also switched the prior mean from 1 (random-walk prior, correct for *levels*) to 0, since FRED-MD series are already transformed to near-stationarity. 2. **Factor model instability**: the original implementation iterated a 1-step model forward 12 times to get an h=12 forecast; any autoregressive coefficient slightly above 1 explodes exponentially under iteration, producing RMSFE outliers as large as 246×. Replaced with the paper's actual specification (Eq. 6–7): a **direct** regression of y_t on y_{t-h} and f_{t-h}, one regression per horizon, no iteration.

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3. Scope reduction (Stage 1 guidelines explicitly permit this)

> *"Exact reproduction of the reported numbers is not required, particularly when datasets, APIs, hardware, or proprietary models are unavailable."*

Dimension	Paper	Ours	Why
Series	120	60	Halves compute; category-proportional subset (output, labor, housing, money, rates/FX, prices, stock market)
Re-estimation frequency	Monthly (~456 origins)	Quarterly (48 origins)	Single biggest runtime lever — same expanding-window mechanic, just fewer origins

Dimension	Paper	Ours	Why
Evaluation window	Jan 1985–Dec 2022	2010–2022	Tractable on CPU-only hardware (no A6000 GPU), still a genuine out-of-sample test including a real regime change (COVID)
Horizons computed	h = 1..12	h = 1, 3, 6, 12	Matches exactly what the paper's own tables report
Econometric benchmarks	BVAR, Factor, NNAR, BART	BVAR, Factor only	NNAR/BART are the two most compute-heavy per-origin, and not required inputs to the hybrid
BVAR prior	Marginal-likelihood-optimized lambda + sum-of-coefficients + single-unit-root dummy priors	Fixed lambda=0.2, symmetric Minnesota shrinkage only	Simpler closed-form version; validated to reproduce the paper's qualitative pattern (see §4)
Factor model lags	BIC-selected	Fixed at 2	Simplification; the *direct-horizon* regression structure (the part that matters for stability) matches the paper exactly
TSFMs	6, zero-shot, full history	Same 6, zero-shot; Moirai capped at 200-obs context	CPU speed — Chronos/TimesFM cap context internally regardless
Fine-tuning	Tested (Appendix E.3), found marginal	Skipped	Paper's own finding is the citable justification
Chronos variant	chronos-t5-base (autoregressive sampling)	chronos-**bolt**-small	The T5-based model took ~13 min/origin on CPU (~10+ hrs for the full loop); Bolt is Amazon's own faster successor (direct multi-quantile regression, not token-by-token sampling) at a comparable size class — ~2000x faster, not a scope cut
Time-GPT querying	1 API call per (date, series) pair	1 API call per date, covering all 60 series (long-format batch)	Cut runtime from ~5.4 hours to ~4 minutes; Nixtla's API natively supports multi-series batches, so this changes nothing about what's being computed
Statistical significance testing	Diebold-Mariano per series (Appendix E.1)	Not done	Time constraint; RMSFE point comparison still gives a real signal
Persistence/domain-heterogeneity analysis (§7.1–7.4)	Done	Not attempted	Stretch goal beyond Stage 1's requirements

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4. Results

4.1 Full comparison table — median RMSFE vs AR(1)

Pre-COVID (2010–2019 origins) — clean comparison, mirrors the paper's own choice to exclude the pandemic period from its main figure:

Model	h=1	h=3	h=6	h=12
BVAR	1.000	1.055	1.007	0.997
Factor-AR	0.981	0.990	0.994	1.002
TimesFM	**0.965**	**0.981**	**0.984**	0.997
Chronos	0.991	0.987	0.986	0.999
Time-GPT	0.980	0.985	0.988	0.998
Moirai	1.032	1.025	1.014	1.005
TTM	1.072	1.011	1.000	1.000
LagLlama	1.079	1.093	1.068	1.014
Hybrid (BVAR+TimesFM residuals)	0.989	1.050	1.008	0.998

Full window (2010–2022, includes COVID) vs. COVID-only subset (2019-06 onward) — shows model fragility to the regime break:

Model	Full h=1	COVID-only h=1	Full h=12	COVID-only h=12
BVAR	1.332	**1.564**	1.108	**1.873**
Factor-AR	1.507	**1.897**	1.029	1.096
TimesFM	0.983	0.973	1.000	1.011
Chronos	0.996	1.022	1.002	1.000
Time-GPT	0.987	0.981	0.999	1.002
Hybrid	1.331	1.578	1.111	1.880

(Full precision numbers and std/min/max in `output/rmsfe_summary.csv`; boxplots in `output/rmsfe_boxplots.png`.)

4.2 Interpretation

TSFM ranking replicates the paper closely. Our best-to-worst ordering — TimesFM ~ Chronos ~ Time-GPT (all beat AR(1) at every horizon) > Moirai (roughly at par) > TTM ~ LagLlama (worse than AR(1)) — matches the paper's own ranking (their best three: Moirai-large, Chronos, TimesFM; their worst two: TTM-enhanced, Time-GPT) with one clear divergence, discussed below.

BVAR/Factor models beat AR(1) pre-COVID, as the paper found, but their performance is *not* stable across regimes — the striking finding here is how badly they break down once COVID enters the sample (RMSFE up to 1.87 vs AR(1) at h=12!), while every TSFM stays within a few percent of 1.0 throughout. This is a real-data confirmation of the paper's own conclusion (§8): *"their [TSFMs'] relatively stronger performance in the post-Covid-19 period suggests an advantage in handling structural breaks."* Our simplified BVAR (no stochastic volatility, no COVID-specific outlier handling like the

paper's own Appendix's cited Carriero et al. 2022 approach) makes this fragility even more visible than in the paper.

Time-GPT is our biggest divergence from the paper. The paper found it one of the two weakest TSFMs (median 1.03–1.08); here it's one of the *best* (0.98–0.99). Most likely explanation: Time-GPT is the only TSFM served via a live, continuously-updated hosted API rather than a frozen local checkpoint — we are almost certainly querying a materially different (newer) model than the one benchmarked in the paper's 2024 evaluation. This is exactly the kind of "differences should be discussed and justified" case Stage 1 asks for.

The hybrid partially replicates the paper's benefit, but not fully. It improves on plain BVAR at h=1 (1.000->0.989) and clearly reduces the tail risk during COVID is inherited rather than fixed (Hybrid tracks BVAR's 1.56–1.87 blowup almost exactly) — because the hybrid's forecast is *anchored* to BVAR's own point forecast plus a residual correction, it cannot fully escape a badly-anchored base forecast during a genuine regime break; the residual correction only fixes the *linear, learnable* part of BVAR's error, not a wholesale level shift. It also never beats standalone TimesFM at any horizon here, unlike the paper. Our best explanation: the paper's BVAR (fully tuned, with sum-of-coefficients and single-unit-root priors) is already a *strong* base model in its own right, so hybridizing a strong base with a TSFM correction can exceed a good TSFM alone; our simplified BVAR is comparatively weaker, capping the hybrid's ceiling below standalone TimesFM.

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5. Conclusion

The core qualitative finding of the paper — TSFMs are competitive with but not clearly superior to well-specified econometric models, and the two are complementary enough that combining them can help — replicates at our reduced scale. The TSFM performance ranking matches closely; the divergence on Time-GPT and the more limited hybrid gain are both explainable by concrete implementation differences (hosted API drift, a less richly-specified BVAR prior) rather than methodological errors, and both are documented above per Stage 1's requirement to discuss and justify any differences from the original paper.